

Mutual Fund Performance Analytics

Project Title

Mutual Fund Performance Analytics Using Python

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Objective

The objective of this project is to analyze and evaluate the performance of mutual funds using financial analytics techniques. The project focuses on measuring returns, risk-adjusted performance, benchmark comparison, and fund ranking for investment analysis.

Business Problem

Investors often struggle to compare multiple mutual funds effectively because:

- Returns alone are not sufficient
- Risk differs across funds
- Benchmark comparison is necessary
- Volatility impacts investment decisions

This project solves the problem by creating a data-driven performance analytics system.

Project Goals

- Analyze 40 mutual fund schemes
- Compute return metrics
- Measure risk-adjusted performance
- Compare funds against benchmarks
- Build a fund ranking system
- Generate investment insights

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy
- Jupyter Notebook

Dataset Overview

Datasets Used

1. Fund Master Dataset

Contains:

- Fund names

- AMFI codes
- Expense ratio
- Fund category

2. NAV History Dataset

Contains:

- Daily NAV values
- Historical fund performance
- Date-wise observations

3. Benchmark Dataset

Contains:

- NIFTY50 index data
- NIFTY100 index data
- Benchmark closing prices

Importance of Data

The datasets help evaluate:

- Historical performance
- Volatility
- Benchmark tracking
- Risk-return relationship

Data Preparation Steps

- Removed duplicates
- Handled missing values
- Converted dates into datetime format
- Standardized column names
- Sorted datasets

Daily Return Analysis

Daily Return Formula

Daily return measures the percentage change in NAV between two consecutive days.

Formula:

$$\text{daily_return} = \text{NAV}(t) / \text{NAV}(t-1) - 1$$

Purpose

Daily returns are important because they:

- Measure short-term performance
- Help calculate volatility
- Form the base for risk metrics

Implementation

Used:

- `groupby()`
- `pct_change()`

Distribution Analysis

A histogram was created to:

- Check return distribution
- Detect outliers
- Understand volatility behavior

Insight

Most returns clustered near zero, indicating normal market movement with occasional extreme fluctuations.

CAGR Analysis

What is CAGR?

CAGR (Compound Annual Growth Rate) measures annualized investment growth over time.

Formula

$$\text{CAGR} = (\text{Ending NAV} / \text{Starting NAV})^{(1/n)} - 1$$

Where:

- n = number of years

Calculations Performed

- 1-Year CAGR
- 3-Year CAGR
- 5-Year CAGR

Importance

CAGR helps:

- Compare long-term performance
- Measure investment growth consistency
- Identify high-performing funds

Observation

Funds with consistently high CAGR showed stronger long-term performance and stability.

Sharpe Ratio Analysis

What is Sharpe Ratio?

Sharpe Ratio measures return earned per unit of total risk.

Formula

$$\text{Sharpe Ratio} = (R_p - R_f) / \sigma$$

Where:

- R_p = portfolio return
- R_f = risk-free rate
- σ = standard deviation

Risk-Free Rate Used

6.5% RBI repo rate proxy

Importance

Sharpe Ratio helps:

- Compare funds fairly
- Evaluate risk-adjusted returns
- Identify efficient portfolios

Interpretation

- Higher Sharpe Ratio = Better performance
- Lower ratio = Higher risk for same return

Result

Top-performing funds demonstrated superior risk-adjusted returns.

Sortino Ratio Analysis

What is Sortino Ratio?

Sortino Ratio measures return relative to downside risk only.

Formula

Sortino Ratio = $(R_p - R_f) / \text{Downside Deviation}$

Difference from Sharpe Ratio

- Sharpe uses total volatility
- Sortino uses only negative volatility

Importance

Investors are more concerned about downside losses than positive fluctuations.

Benefits

- Better downside risk measurement
- More accurate for investment analysis
- Useful in portfolio management

Observation

Funds with higher Sortino ratios handled negative market movements more effectively.

Alpha and Beta Analysis

Beta

Beta measures sensitivity of fund returns compared to benchmark returns.

Interpretation

- $\text{Beta} > 1 \rightarrow$ More volatile than market
- $\text{Beta} < 1 \rightarrow$ Less volatile
- $\text{Beta} = 1 \rightarrow$ Similar to benchmark

Alpha

Alpha measures excess return generated beyond benchmark expectations.

Method Used

OLS regression using `scipy.stats.linregress()`

Benchmark Used

NIFTY100 index

Importance

Alpha and Beta help investors understand:

- Market sensitivity
- Fund manager performance
- Benchmark outperformance

Observation

Positive alpha funds outperformed the benchmark index consistently.

Maximum Drawdown Analysis

What is Maximum Drawdown?

Maximum Drawdown measures the largest decline from peak NAV to lowest NAV.

Formula

$\text{Drawdown} = \text{NAV} / \text{Running Max} - 1$

Purpose

Used to:

- Measure downside risk
- Evaluate capital loss
- Understand worst-case performance

Importance

Lower drawdown indicates:

- Better capital protection
- Lower risk exposure
- Stronger portfolio resilience

Observation

Funds with smaller drawdowns showed greater stability during market declines.

Fund Scorecard System

Objective

Create a composite ranking model for all mutual funds.

Scoring Weights

Metric	Weight
3Y CAGR Rank	30%
Sharpe Rank	25%
Alpha Rank	20%
Expense Ratio Rank	15%
Max Drawdown Rank	10%

Normalization

Scores were normalized between 0–100.

Importance

The scorecard provides:

- Easy comparison
- Data-driven ranking
- Investment decision support

Outcome

Top-ranked funds demonstrated:

- Strong returns
- Better risk management

- Lower expense burden

Benchmark Comparison & Conclusion

Benchmark Comparison

Top 5 funds were compared against:

- NIFTY50
- NIFTY100

Tracking Error

Tracking error measures deviation from benchmark returns.

Formula

Tracking Error = $\text{Std}(\text{Fund Return} - \text{Benchmark Return}) \times \sqrt{252}$

Importance

Tracking error helps:

- Measure benchmark consistency
- Identify active management impact
- Evaluate portfolio alignment

Final Conclusion

The project successfully:

- Evaluated 40 mutual fund schemes
- Measured return and risk metrics
- Compared funds against benchmarks
- Developed a fund ranking model

Key Learnings

- Risk-adjusted metrics are more important than raw returns
- Benchmark comparison is critical in fund analysis
- Data analytics improves investment decision-making

Future Enhancements

- Power BI dashboard integration
- Real-time NAV tracking
- Machine learning return prediction

- Streamlit web application
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